SECTION 1

The Curse of Dimensionality

- **Sparse high-D space:** Points cluster near hypercube borders; random pairs are far apart, so neighbors-based learning extrapolates unreliably.
- **Overfitting risk:** More features raise the chance of fitting noise unless the training set grows exponentially with dimension.
- **When to reduce:** Try strong baselines on full data first; compression trades information for speed and simpler pipelines.
- **Visualization motive:** Mapping to two or three dimensions reveals clusters and communicates structure to non-specialists.

SECTION 2

Projection and Manifold Learning

- **Projection:** Many datasets lie near a low-dimensional subspace; orthogonal projection onto that subspace drops dimensions with modest loss.
- **Swiss roll caveat:** A naive projection squashes curved manifolds; unfolding requires manifold learning, not a single linear drop.
- **Manifold hypothesis:** Real data often concentrates on a lower-dimensional manifold embedded in feature space.
- **Task dependence:** A simpler boundary in 3D may become complex after unrolling, so reduction is not always beneficial for accuracy.

SECTION 3

Principal Component Analysis

- **Variance objective:** PCA finds orthogonal directions that preserve maximum variance; the first axes minimize reconstruction error.
- **SVD link:** Centered data decomposed via SVD yields principal component directions in `components_`.
- **`PCA` in Scikit-Learn:** `fit_transform` centers data and projects to `n_components` dimensions.
- **Explained variance:** `explained_variance_ratio_` shows how much variability each PC retains; cumulative sums guide `n_components=0.95` selection.
- **Compression:** MNIST at 95% variance shrinks features dramatically with small reconstruction error via `inverse_transform`.
- **Randomized and incremental PCA:** `svd_solver="randomized"` speeds large sparse problems; `IncrementalPCA` streams mini-batches or memory-mapped arrays.

SECTION 4

Kernel PCA

- **Nonlinear embeddings:** The kernel trick maps data implicitly to a high-D space where linear PCA captures nonlinear structure.
- **`KernelPCA`:** Choices such as RBF can separate clusters or unroll rolls where linear PCA fails.
- **Hyperparameter search:** Supervised pipelines with `GridSearchCV` can tune kernel and gamma for downstream accuracy.
- **Reconstruction:** Feature-space pre-images are approximate; `fit_inverse_transform=True` learns a mapping to score reconstruction error in input space.

SECTION 5

Locally Linear Embedding

- **Manifold unfolding:** LLE preserves local linear combinations of neighbors when embedding in fewer dimensions.
- **`LocallyLinearEmbedding`:** `n_neighbors` sets locality; good for unrolling swiss rolls but can distort global distances.
- **Two-step procedure:** First solve for neighbor weights, then find low-D coordinates that reproduce those weights.
- **Scalability:** Cost includes $O(m^2)$ terms, limiting use on very large sets without approximations.

SECTION 6

Other Dimensionality Reduction Techniques

- **Random projections:** Johnson–Lindenstrauss guarantees preserve distances with surprisingly few random dimensions.
- **MDS and Isomap:** Preserve pairwise distances or geodesics on neighborhood graphs.
- **t-SNE:** Favors local neighborhood structure for visualization, often separating digit classes in 2D plots.
- **LDA:** Supervised method projecting onto axes that maximize class separation, useful before classifiers such as SVMs.